

Operation Blackout

Consolidated technical investigation dossier: from first contact to confirmed compromise

CALLSIGN	rupesh
INVESTIGATOR	Rupesh Malpani
CASE	HX-1977 / Operation Blackout
SCOPE	Main chain, both branches, finale, epilogue and Independent Arcade
STATUS	19 of 19 challenges solved, verified and documented
ISSUED	27 September 2026 / Asia-Calcutta (IST)
DOCUMENT	HX-1977-FINAL-DOSSIER / version 1.0

Transmission record

This dossier consolidates the challenge-level write-ups retained in the repository. It records the reasoning chain, evidence pivots, execution details, validation controls, tooling, AI assistance and final proof for each solved stage. The stage write-ups remain the detailed technical record; this document is the navigable case-level account.

Field	Recorded value
Callsign / player ID	rupesh — confirmed in the authenticated case state and Tunnel bootstrap evidence
Case reference	HX-1977
Submission window	Flags close 20:00 IST; dossier closes 22:00 IST; 27 September 2026
Covered stages	Nine main-chain/branch/finale stages plus all ten Independent Arcade challenges
Repository convention	challenges/<category>/<challenge>/{writeup.md,assets,analysis,scripts}
Evidence policy	Original acquisitions retained under <code>assets/</code> ; transformations written under <code>analysis/</code> ; reproducible helpers retained under <code>scripts/</code> .
Validation standard	A readable candidate or plausible theory was not accepted as proof. Proof required hashes, format checks, cryptographic verification, deterministic replay, live service acceptance, held-out evaluation or case-board acceptance as appropriate.

19 / 19	10	4,735
DOCUMENTED SOLVES	INDEPENDENT ARCADE SOLVES	FINAL RECORDED POINTS

Evidence handling note. Authentication was performed manually by the participant. Browser automation operated only inside the authorized CTF session. Credentials and cookies were not printed into the write-ups. Live interactions were limited to the challenge interfaces and supplied endpoints.

CONTENTS

00	Executive overview	chain	S2	The Insider	Web
01	Method and tool disclosure	controls	S3A	The Evidence	Forensics
S0	Recon	OSINT	S4A	The Cipher	Cryptography
S1	The Portal	Web	S3B	The Tunnel	Network

S4B	The Implant	Reversing	RC	Required coverage matrix	19 challenges
S5	Nightjar's Nest	Pwn	OL	Open leads	status
EP	Epilogue	Detection	A	Evidence and script index	audit
IA	Independent Arcade	10 puzzles			

Attack chain reconstructed

The investigation followed the application’s handover graph and closed both blue and red branches before the finale.



OUTCOME LEDGER

Stage	Finding / exploit	Acceptance proof	Result
S0 Recon	Provenance reconstruction and exact compact-manifest hash	/recon/verify → ok:true	flag{recon_e8f25be55df46671}
S1 The Portal	TOCTOU between approved preview and mutable delivery reference	Restricted version delivered after direct read returned 403	flag{portal_1285d111fcc3511d}
S2 The Insider	Case-sensitive subjects counted as two reviewers for one employee	Two signed tickets accepted; admin release succeeded	flag{insider_5594d69fa2cb5e59}
S3A The Evidence	Timing-channel recovery across two independently encoded shares	Both lane CRCs, XZ recovery and verifier acceptance	flag{evidence_23c52b9d18c24656}
S4A The Cipher	AES-GCM nonce reuse enabled ciphertext and tag forgery	Live validator authenticated forged release record	flag{cipher_1c06b6f19064bf9d}
S3B The Tunnel	Partially bound journal cursor permitted generation rebinding	Fresh TLS exchange exported historical device key	flag{tunnel_af287b68618f8777}
S4B The Implant	Unsigned module reached unapproved destination; memory retained credential	Unique causal chain plus 192/192 authenticated contexts	flag{implant_341adf156bb91b3d}
S5 Nightjar’s Nest	Queued raw pointer survived revision free; UAF became arbitrary R/W	Live receipt response and board acceptance	flag{nest_83cacdb429a88413}
EP Epilogue	Cross-domain behavioral correlation for unauthorized DNS transfer	35/35 training, 0 FP; one successful held-out evaluation	flag{epilogue_f5e952d7d3c3fc64}

CROSS-STAGE EVIDENCE LOGIC

The chain was not nine isolated flag searches. Each solve produced a receipt, archive credential, device key or factual model consumed by a later stage. The decisive pattern was consistent across the operation: identify the authoritative state, preserve exact bytes, model how state changes, form a narrow hypothesis, test it locally, then use the live endpoint only as final confirmation.

Confirmed compromise. The combined record demonstrates a progression from provenance and authorization weaknesses, through covert export and cryptographic/protocol failures, to memory-resident state recovery, arbitrary process memory access and a transferable behavioral detection. Every stage listed above has independent acceptance evidence.

INDEPENDENT ARCADE OUTCOME

The ten arcade puzzles were independent of the handover chain and contributed 1,475 base points. Each was reconstructed from its original authorized artifact, solved with a retained script, and rechecked through its dedicated answer verifier. Together with the 3,010 documented stage points and 250-point both-branch bonus, the authenticated final state recorded **4,735 points and 19/19 solved**.

Evidence standard and tools used

OPERATING METHOD

1. **Preserve.** Retain supplied archives and binaries; calculate SHA-256 before analysis.
2. **Separate.** Keep primary evidence in `assets/`, derived bytes in `analysis/`, and repeatable logic in `scripts/`.
3. **Correlate.** Use timestamps, transaction semantics, identifiers, checksums and cryptographic bindings instead of trusting a single attractive clue.
4. **Disprove.** Retain failed controls—403 responses, generic verifier failures, CRC failures, protocol errors and classifier false positives—to show why the accepted interpretation is stronger.
5. **Verify.** Require byte-exact regeneration or an authoritative endpoint response before calling a stage solved.

TOOL DISCLOSURE

OpenAI Codex	Used as a coding, reverse-engineering and verification copilot: inventorying files, drafting parsers and solvers, comparing binary layouts, translating rules into deterministic scripts, testing hypotheses, organizing evidence and editing the write-ups. Codex output was never treated as proof by itself.
Playwright MCP	Used to communicate with the browser after manual login: navigate the case board, capture UI/accessibility state, inspect public contracts and responses, download designated acquisitions, issue same-origin requests, and submit already-validated flags. It did not create or bypass authentication and did not disclose session cookies.
Python 3	Dependency-light forensic parsing and solvers: RIFF metadata, transaction journals, FAT12, SQLite/WAL, PCAP/DNS streams, timing channels, ECC/CRC, ELF/core structures, protocol frames, GF(2 ¹²⁸) arithmetic and local detector evaluation.
Node.js	Authenticated browser/CDP helpers, WebCrypto/OpenSSL-compatible routines, core-state reconstruction, acquisition helpers and controlled submission scripts.
Browser DevTools / CDP	Used when an existing Playwright profile was occupied, allowing safe attachment to the already authenticated challenge browser for authorized downloads and response inspection.
SQLite	Used both directly and through Python's standard library to reconstruct committed state, restore a WAL relationship, and execute the final read-only detection query.
Static analysis	ELF headers, symbols, strings and x86-64 disassembly were inspected with retained local scripts and platform utilities. Downloaded executables were not run on the host unless they were purpose-built solver helpers.
Hashing / crypto	PowerShell <code>Get-FileHash</code> , Python <code>hashlib</code> , Node <code>crypto</code> , HMAC, AES-CTR/GCM, CRC-16/32, SHA-256 and exact byte comparisons provided integrity and authenticity controls.
Official web sources	For Recon only, official NASA and JPL sources named by the evidence were used to resolve historical fields. The source context—not merely factual plausibility—determined the accepted identifier.

HOW AI OUTPUT WAS VALIDATED

AI-assisted activity	Required independent check
OCR, visual comparison or audio triage	Original image features, acquisition notes, RIFF metadata and linked mail records
Transaction/state reconstruction	Deterministic replay with applied/rejected/rollback counts and exact committed output
Binary structure inference	CRC checks, pointer-range checks, disassembly, byte counts and known-good traces
Cryptographic interpretation	Recomputed source tags/HMACs, successful decryption/inflation and validator acceptance
Exploit implementation	Fresh session values, sanity assertions, service proof and case-board acceptance
Detection query	Complete training confusion matrix and one independent held-out evaluation

Known AI errors were retained. Examples include choosing the wrong historical identifier, assuming hex-line rather than binary network framing, misreading a final HMAC argument, swapping event status/flags, taking the wrong token slice, and adding the wrong command delimiter. Each error was corrected by returning to primary evidence or a deterministic validator—not by hiding failed attempts.

Recon — disputed custody transfer

SOLVED

PROVENANCE

TRANSACTION REPLAY

The first stage required binding a badge holder to the correct archive, replaying custody transactions, identifying the north-archive photograph and building an exact hash manifest from approved historical sources.

APPROACH AND EVIDENCE TRAIL

The badge identified **Maya Sen / HX-4821**. Audible voicemail content was generic, but the WAV RIFF metadata carried mailbox=archive-17; record=call-2204. Mail m-220, whose subject was call-2204, bound Maya to asset B04-21b0096b; follow-up mail rejected revision a and approved revision c.

The custody journal was replayed with optimistic transaction semantics. Only commits whose base generation and owner matched current state were applied. The final state was generation **39**, carrier f27f23a9, location north-archive; 39 commits applied, 14 were rejected, 20 rolled back and one remained unfinished.



Figure S0-1. Preserved badge evidence identifying Maya Sen and employee ID HX-4821.



Figure S0-2. Corridor evidence: left door/green triangle defines the north archive.



Figure S0-3. North archive interior, corroborated by green-triangle cabinet, red pipe and yellow trolley. Original SHA-256: c287b6b0...d1cdc90.

EXECUTION RECORD

```
# Reproduce committed custody and exact manifest digest
python replay_custody.py

fields = ["B04-21b0096b",39,"f27f23a9",
         "c287b6b0bdb55cc0f5c7d4ec3b4bfd093019bea82e0ef354336438430d1cdc90",
         "1977-07-29","1977-0800",60,55,12,14,"Timothy Ferris"]
manifest = sha256(json.dumps(fields,separators=(",",":")).encode()).hexdigest()
# 4f3bc799d152281d139bac1d2394eecb5656f4da1d7752ca76d8556d4e117c97
```

REASONING AND PROOF

The evidence-approved JPL page used release identifier 1977-0800. An initial use of the legitimate but wrong-context NASA identifier 77-159 failed, proving that source scope was part of the challenge. The corrected compact array was submitted with revision c.

VERIFIED RESPONSE

```
POST /recon/verify
{"revision":"c","manifest":"4f3bc799d152281d139bac1d2394eecb5656f4da1d7752ca76d8556d4e117c97"} →
{"ok":true,"flag":"flag{recon_e8f25be55df46671}"}
```

AI/tool use: Codex assisted with image comparison, RIFF parsing, journal replay and source correlation; Playwright MCP submitted the already-derived manifest from the authenticated origin. All visual and historical values were checked against retained artifacts and approved official pages.

Detailed record: [challenges/osint/recon/writeup.md](#)

The Portal — restricted delivery

SOLVED

TOCTOU

AUTHORIZATION

A collection author could not directly read the restricted incident record. The exploit instead separated the approved object from the object eventually delivered.

EVIDENCE AND HYPOTHESIS

The API contract, database model and support logs showed that a preview stored an immutable version and digest, but an export job retained a mutable `reference_id`. Delivery reopened the reference after pause/resume. The control request to the restricted content endpoint returned **HTTP 403**, demonstrating that knowing its identifier was not enough.

EXECUTION RECORD

1. Create a collection containing allowed document `brief-d178a81cb6`.
2. Import it, create `preview-9`, approve it as `approval-10`, and queue `job-11`.
3. Pause the job at cursor 0.
4. Patch `collection-6` revision 1 so its member becomes restricted `incident-d178a81cb6`.
5. Resume and dispatch the original job without a new preview or approval.

```
PATCH /portal/collections/collection-6
{"member":"incident-d178a81cb6","revision":1}

POST /portal/jobs/job-11/resume
POST /portal/jobs/job-11/dispatch
GET /portal/jobs/job-11/download
```

PROOF AND PRESERVATION

The completed job retained `approval-10` but reported `delivered_version: incident-d178a81cb6:1`. The downloaded bytes contained the restricted incident and acquisition receipt. They were preserved as `portal-job-11-delivery.txt`, SHA-256 `3640a514...a59c640`.

VERIFIED RESPONSE

```
direct restricted read → HTTP 403 {"detail":"document read denied"} approved preview → allowed brief
delivered_version → incident-d178a81cb6:1 download receipt → flag{portal_1285d111fcc3511d}
```

AI/tool use: Playwright MCP navigated the workspace, inspected network responses and executed the documented state transitions. Codex correlated `models.sql`, release notes and the delivery log, then retained the response hash. The exploit was verified by the version mismatch and byte-exact download, not by UI appearance.

Detailed record: [challenges/web/the-portal/writeup.md](#)

The Insider — emergency release record

SOLVED

IDENTITY CANONICALIZATION

QUORUM

The effective workflow state had to be replayed before testing authorization. Policy P19 then counted two subject strings as two reviewers even though they belonged to one human employee.

EVIDENCE TRAIL

Deterministic replay of `workflow.db` rejected transactions containing failed compare-and-swap operations, even when a later row said COMMIT. Effective state was case `0144b1151f1c`, revision 3, request `042e4765d649df`, policy P19, root `751a4690aa0a6da49e4c`, status `pending-quorum`.

P19 joined directory rows on employee number but counted case-sensitive subjects. The federated and on-prem records had subjects `rv-46e9334a6c` and `RV-46e9334a6c`, while both had employee number `46e9334a6c`. The policy counted two principals where only one person existed.

EXECUTION RECORD

```
python replay_workflow.py
# revision=3 policy=P19 status=pending-quorum

POST /insider/recover {handover, manifest}
POST /insider/review {handover, object_id:"a99cc695bf06389f3d5a"}
POST /insider/review {handover, object_id:"ae95670c58b2db7cc59a"}
POST /insider/approve {handover, review_ticket, manifest} # twice
POST /insider/admin {handover, manifest}
```

PROOF

No ticket was forged. The server issued two valid, state-bound review tickets and accepted both. Their principals differed only by case; their employee identity was the same.

VERIFIED RESPONSE

```
review ticket 1 → principal rv-46e9334a6c review ticket 2 → principal RV-46e9334a6c employee number →
46e9334a6c (same human) approve #1 → {"accepted":true} approve #2 → {"accepted":true} admin release →
{"ok":true,"flag":"flag{insider_5594d69fa2cb5e59}"}
```

AI/tool use: Codex wrote and checked the replay logic, compared policies and joined identities. Playwright MCP retrieved the acquisition, requested signed tickets and issued the documented requests after manual login. Proof rested on committed state, signed ticket payloads and the successful final release.

Detailed record: [challenges/web/the-insider/writeup.md](#)

The Evidence — gateway acquisition

SOLVED

FIRST BLOOD

TIMING CHANNEL

A recovery document crossed the boundary inside historian timing. The attribution required committed workstation state, clock correction, DHCP history, network corroboration and two independently decoded shares.

ATTACK CHAIN AND FINDINGS

The workstation image contained RECOV.DB and its WAL; restoring the expected basename relationship let SQLite apply committed frames. The export display time 2026-09-27 14:14:28 was UTC+05:30 and 137 seconds fast, yielding 2026-09-27T08:42:11Z. DHCP history mapped reception-17 to 10.27.4.18 at that time, while .44 belonged to an approved ops host.

Because secure_delete=OFF, deleted R9CF profiles survived in unallocated SQLite bytes. Profile CRC-16, generation 24955 and calibration captures selected the active profiles. Timestamp-rank permutations yielded two 233-byte shares; each passed its stored CRC-32. XOR exposed an XZ stream whose decompressed 177-byte JSON matched the host task and owner.

EXECUTION RECORD

```
python scripts/triage_evidence.py
python scripts/extract_fat_files.py
python scripts/inspect_reception_db.py
python scripts/extract_dns_streams.py
python scripts/analyze_historian.py
python scripts/decode_r9_profiles.py
python scripts/recover_rank_symbols.py
python scripts/locate_r9_frames.py
python scripts/recover_r9_lanes.py
```

VERIFIED RESPONSE

```
LANE 0 stored_crc32=8d01014f calculated_crc32=8d01014f LANE 1 stored_crc32=dcdf35b0 calculated_crc32=dcdf35b0
combined_size=233 xz_offset=13 document_size=177
document_sha256=4484648495b0dab0b48e533c638cdc5224f5d86d7bedbf8fe4cf88283681de verifier →
{"ok":true,"flag":"flag{evidence_23c52b9d18c24656}"}
```

FINAL EVIDENCE TUPLE

```
{"source":"10.27.4.18","task":"db06efd5a7549b9e",
"exported_utc":"2026-09-27T08:42:11Z",
"document_sha256":"4484648495b0dab0b48e533c638cdc5224f5d86d7bedbf8fe4cf88283681de"}
```

AI/tool use: Codex assisted with FAT/PCAP/SQLite parsers, ELF interpretation and inverse timing transforms; browser automation downloaded the designated bundle and submitted one verified tuple. Every transformation was bounded by hashes, calibration, sync, ECC, CRC, XZ structure and server acceptance.

Detailed record: [challenges/forensics/the-evidence/writeup.md](#)

The Cipher — forged recovery journal

SOLVED

AES-GCM

NONCE REUSE

Packet-arrival phase hid a wrapped journal. Two equal-length records reused one AES-GCM nonce, exposing both the counter stream and enough GHASH structure to forge an authenticated release command.

EVIDENCE TRAIL

Of 37,496 UDP JSON messages, 23 sensors sent 900 packets while sensor F-2F9E sent 16,796. Computing collector time minus transmitter time separated ordinary ± 12 ms samples from ~ 60 ms delayed samples. At bit offset 364 the stream contained TRP2. Registration-order calibration bytes and the Evidence archive key derived AES-CTR wrap key a018019148d17832c2965bb14c49a239; strict zlib inflation produced a 19-record journal.

Records 7 and 14 reused nonce 18a36dba1453cb10180bc217. Retained activity plus the 48-byte command ABI reconstructed both plaintexts. Their tag difference and first-block ciphertext difference gave H^4 ; finite-field inversion recovered $H=518ad05ada2c651572c5817c23a5d2f8$. Both original tags were recomputed before forging.

EXECUTION RECORD

```
python scripts/analyze_cipher_pcap.py
python scripts/recover_cipher_transport.py
node scripts/recover_cipher_journal.js
python scripts/forge_cipher_release.py

nonce      = 18a36dba1453cb10180bc217
ciphertext = 1b944dd2b1309d9598ea8bb1be7972055ce673a951e8631f53cfef01b0f02656ff90f8278db07f42b79bb6f464d56398
tag        = 38ac1db1c473c0c8792867714c3da7f9
```

PROOF

FAILED CONTROL / CORRECTIVE EVIDENCE

First delimiter hypothesis $\rightarrow$ {"ok":false} Cause: read context incorrectly ended with a pipe; fixed-width ABI actually used spaces.

VERIFIED RESPONSE

POST /cipher/dispatch original source tags recomputed $\rightarrow$ both match forged plaintext $\rightarrow$ exact 48-byte action=release command validator $\rightarrow$ {"ok":true,"flag":"flag{cipher_1c06b6f19064bf9d}"}

AI/tool use: Codex generated the dependency-free PCAP parser, unwrap logic and $GF(2^{128})$ implementation. Playwright MCP served only as authenticated same-origin transport. One official hint was redeemed (10% deduction); its direction was to compare authenticated journal records with retained plaintext. Exact tag reproduction and validator acceptance controlled the final result.

Detailed record: [challenges/cryptography/the-cipher/writeup.md](#)

The Tunnel — retained device journal

SOLVED

PROTOCOL REVERSING

CURSOR REBINDING

A journal cursor mixed an object reference, embedded generation and session binding without cryptographically binding those meanings together. Recombining valid parts in a fresh session selected historical committed material.

EVIDENCE AND REASONING

The stripped client, shared library, ABI and two laboratory traces established raw frame structure, CRC-16, HMAC inputs and command sequencing. A cursor was 8-byte object reference | 4-byte generation | 8-byte session binding. The service checked current outer and embedded generations, but did not bind the historical object reference to its original generation.

In a fresh TLS session, the retained state-3 cursor was 69bd92377a533352|d83c0000|b6b1d1410fc841c3; current generation was d83c0100. Replacing only bytes 8–11 preserved the historical reference and live session tail while satisfying the generation checks.

EXECUTION RECORD

```
python challenges/network/the-tunnel/scripts/solve_tunnel.py --verify
# both laboratory captures regenerated byte-for-byte through command 0x23

python -u challenges/network/the-tunnel/scripts/solve_tunnel.py
# OPEN → BEGIN → SCAN → SUSPEND → EXPORT → FINISH

rebound = journal_cursor[:8] + pack("<I", open_generation) + journal_cursor[12:]
```

VERIFIED RESPONSE

```
EXPORT metadata → historical ref 69bd92377a533352 / generation d83c0000 device key →
64c5d828bac56fe9037c4a188d21652c0df5221c6118320ee36541d32c01dcf7 final proof → HMAC(device_key, session_id ||
nonce || SHA256(transcript)) server → {"ok":true,"flag":"flag{tunnel_af287b68618f8777}"}
```

CORRECTIONS AND VALIDATION

Recorded frames could not be replayed because session challenges, HMACs and cursor tails expired. Three failed interpretations produced useful evidence: treating the service as hex-line transport, using historical generation as the outer generation, and using the maintenance credential as the final proof key. Binary framing and System V register analysis corrected these.

AI/tool use: Codex compared traces, translated disassembly, built framing/CRC/HMAC/TLS code and proposed generation tests. The solver verified certificate identity, checked every frame and reproduced known traces offline before one fresh live exchange. Playwright/browser helpers retrieved only authorized bootstrap material.

Detailed record: [challenges/network/the-tunnel/writeup.md](#)

The Implant — interrupted process acquisition

SOLVED

CORE FORENSICS

AES-GCM

All 192 retained contexts authenticated, so cryptographic validity alone could not identify the incident. A CRC-valid circular event ring isolated the one complete unsigned-module and unapproved-destination transfer, which mapped to current context 35.

EVIDENCE TRAIL

The ELF core's virtual addresses were mapped through PT_LOAD segments. Valid RCTX headers led through masked linked-node chains; the Tunnel device key derived per-context authentication material. Parsing the ring by sequence number, not physical position, found a single complete chain:

```
begin(handle)
  → module load(handle, unsigned)
  → connect(handle, unapproved)
  → stage(head)
  → transmit(head)
  → cleanup(handle)
```

Sequences 10637–10651 shared handle 0x000067b2c59e6291; stage/transmit shared head 0x00007f40918b1000. That head matched current header 35, owner 2419674196, generation 3850136661. Its 31-node, 1,505-byte state authenticated with AES-256-GCM.

EXECUTION RECORD

```
python scripts/analyze_implant.py assets/implant-process.core
python scripts/analyze_implant_events.py assets/implant-process.core
node scripts/solve_implant.js assets/implant-process.core \
  64c5d828bac56fe9037c4a188d21652c0df5221c6118320ee36541d32c01dcf7 35
node scripts/submit_implant.js
```

VERIFIED RESPONSE

```
records=1536 valid=1535 invalid=1 completed policy-violating chains=30 completed full-incident chains=1 AUTH
owner=2419674196 generation=3850136661 nodes=31 bytes=1505 credential=a648b8e3fbb05b5a70572e5cae51ec92
endpoint → {"ok":true,"flag":"flag{implant_341adf156bb91b3d}"}
```

CORRECTIVE EVIDENCE

The first parser swapped status and flags; the first token slice included owner and generation. CRC-valid records, the acquisition notes and little-endian parsing exposed both mistakes. The correct token begins at plaintext offset 0x0c, after magic, owner and generation.

AI/tool use: Codex assisted with ELF mapping, disassembly, pointer and crypto formulas, event parsing and hypothesis tests. Browser/CDP automation acquired the authorized files and submitted final evidence. No downloaded executable was executed; all claims were checked against structural CRCs, pointer ranges, chain invariants, GCM tags and the evidence endpoint.

Nightjar's Nest — document spool worker

SOLVED

USE-AFTER-FREE

ARBITRARY R/W

A queued job retained a raw revision pointer without acquiring ownership. undo changed a mutable ID, allowing compact to free the queued revision. Documented render and patch operations then dereferenced attacker-controlled replacement data.

TRIGGER AND PRIMITIVE

```
create doc0,32 → save → replace doc0,32 → queue job0
undo doc0      → saved revision ID mutates
compact doc0   → queued revision freed; job0 remains
create doc1,64 → controlled allocation overlaps stale Revision
queue job1     → patch job1 retargets fake Revision.data
render job0    → arbitrary 32-byte read
patch job0     → arbitrary 32-byte write
```

Rendering the freed object leaked the arena-global pointer at image-relative `0x4a80`, giving the PIE base. The fake revision then targeted globals at `0x4a50` to copy randomized required-delivery state over completed state, and `0x4a90` to follow the heap pointer to the session credential.

EXECUTION RECORD AND PROOF

```
python challenges/pwn/nightjars-nest/scripts/solve_nest.py
# Uses verified TLS, fresh ASLR values and only documented spool operations.
```

VERIFIED RESPONSE

```
example fresh session: PIE base = 0x60b322fd5000 required state = 0x7bd965ee7643401f credential buffer =
0x60b3330042b0 credential = dfa36d5dea71c59d7c89e7eb8e17b6850088bb893ee94f1aec40ef4a55e61d46 service response
= flag{nest_83cacdb429a88413} case board = {"correct":true,"points":600,"first_blood":false}
```

ASLR addresses, state and credential change every session; the solver derives them rather than using the captured example. No shell, filesystem access or code execution was required.

AI/tool use: Codex supported stripped-binary structure recovery, allocator reasoning, protocol implementation and exploit assertions. The result was validated against live, randomized process state and independent case-board acceptance.

Detailed record: [challenges/pwn/nightjars-nest/writeup.md](#)

Epilogue — transferable DNS-transfer detection

SOLVED

FIRST BLOOD

HELD-OUT EVALUATION

The finale required a behavioral detector that survived randomized hosts, IPs, domains, PIDs, hashes, certificates, order and payload. Hard-coded indicators could not pass.

DETECTION LOGIC

The final read-only SQL correlated three independent evidence domains:

1. **Resolver:** normalize collector clocks, resolve address ownership using the active lease, de-duplicate retransmissions, validate CRCs and require one unambiguous complete stream of at least ten chunks.
2. **Endpoint:** normalize module/process clocks, require a recent module load by a process with a recorded parent, and bind the event to the transfer host.
3. **Control state:** at module-event time, require both absence of valid/unrevoked signer trust and absence of an effective exact host/module deployment approval.

The first rule achieved full recall but only 77.8% precision because it treated short repeated deliveries as substantive transfer streams. The minimum complete-stream condition eliminated those benign cases without encoding training identifiers.

EXECUTION RECORD

```
python challenges/detection-engineering/epilogue/scripts/epilogue_analyze.py
# loads training JSON into SQLite, registers nj_* helpers, executes exact SQL

detector bytes=2455 predictions=35 tp=35 fp=0 fn=0
precision=1.000 recall=1.000

# exact query retained at:
challenges/detection-engineering/epilogue/analysis/epilogue-detector.sql
```

PROOF

VERIFIED RESPONSE

```
training corpus → 35 incidents / 35 benign / 0 errors held-out corpus → 240 streams with randomized identities
and state evaluation calls → 1 of 20 POST /epilogue/grade →
{"ok":true,"flag":"flag{epilogue_f5e952d7d3c3fc64}"} case board →
{"correct":true,"points":220,"first_blood":true}
```

AI/tool use: Codex translated the contract into predicates, constructed a local SQLite harness, analyzed false positives and simplified the final SQL. Playwright MCP performed the one authenticated evaluator request and flag submission. No held-out labels were visible; transferability was established only by the independent evaluator.

Detailed record: [challenges/detection-engineering/epilogue/writeup.md](#)

Independent Arcade — ten standalone solves

10 / 10 SOLVED INDEPENDENT ARTIFACTS ANSWER VERIFIED

The arcade challenges did not depend on case handovers and could be solved in any order. Their artifacts were recovered from the authorized puzzle endpoints, preserved under `challenges/arcade/`, and paired with one deterministic solver and one individual write-up each.

Challenge	Category	Points	Recovered answer
Route Zero	Graph search	25	49
Pixel Ledger	Nonogram	75	1110100011110001
Deadline Lab	Scheduling	75	CBAEFD
Double Entry	Accounting	100	30186
Beacon Six	Signal analysis	150	609637
Terminal Echo	Terminal emulation	150	AEYIOAKN
Custodian Records	Secret sharing	200	140
Switchboard	Boolean logic	200	01110011
Device Messages	RSA	300	760422
Afterlife	Cellular automata	200	011000001000101011001111110000000000

Aggregate proof. Re-running `node challenges/arcade/verify_arcade_answers.js` against the authenticated puzzle verifiers returned `ok:true` for all ten answers and reproduced all ten flags. This verification does not resubmit flags to the case board.

Route Zero

DIJKSTRA

DIRECTED GRAPH

Find the minimum directed-route cost from station A to station F.

EVIDENCE AND APPROACH

The artifact contained eight weighted directed edges. Dijkstra's algorithm started at A=0, reached C=19 and B=26, then set E=33 through C. Although B improved D to 41, the E route reached F first at 49.

```
A -> C -> E -> F
19 + 14 + 16 = 49

python challenges/arcade/route-zero/scripts/solve.py
# path=A->C->E->F
# cost=49
```

PROOF AND TOOLS

VERIFIED RESPONSE

```
answer 49 → {"ok":true,"flag":"flag{welcome_fdb3ed9c4339c847}"}
```

Codex implemented the shortest-path checker; all candidate A-to-F paths were also compared manually. Playwright/CDP retrieved the exact JSON and used the official answer verifier. Artifact SHA-256: 813c09da...25ee65e.

Individual record: challenges/arcade/route-zero/writeup.md

Pixel Ledger

CONSTRAINT SOLVING UNIQUE GRID

Resolve a 4×4 nonogram and submit the row-major binary grid.

REASONING

The third row clue [4] forced all cells on. Column 1's [3] then fixed rows 1–3 filled and row 4 empty. Column 4's [2] paired rows 3 and 4. The two split-run columns completed the unique grid:

1	1	1	0
1	0	0	0
1	1	1	1
0	0	0	1

Its columns are 1110, 1010, 1010, and 0011, exactly matching [3], [1,1], [1,1], and [2].

```
python challenges/arcade/pixel-ledger/scripts/solve.py
# exhaustive 2^16 enumeration; exactly one solution
# answer=1110100011110001
```

VERIFIED RESPONSE

answer 1110100011110001 → {"ok":true,"flag":"flag{badge_391d9e9b56cb40a1}"}

Codex encoded the clue evaluator; uniqueness and every clue were independently checked. Artifact SHA-256: 9922f702...0930194.

Individual record: challenges/arcade/pixel-ledger/writeup.md

Deadline Lab

WEIGHTED COMPLETION PRECEDENCE

Order six jobs to minimize weighted completion time while enforcing A before D and B before F.

EXECUTION AND RESULT

With only 720 permutations, exhaustive search was the strongest proof. Invalid precedence orders were rejected; each remaining order accumulated duration and completion × weight. Ties were compared alphabetically as required.

Job	Completion	Weighted cost
C	2	10
B	7	35
A	16	80
E	24	96
F	29	58
D	36	72

```
python challenges/arcade/deadline-lab/scripts/solve.py
# order=CBAEFD
# cost=351
```

VERIFIED RESPONSE

answer CBAEFD → {"ok":true,"flag":"flag{rota_bfc9b384d4620514}"}

Codex implemented exhaustive enumeration and the exact tie-break rule. Artifact SHA-256: a5a0bc48...6e4d3a5.

Individual record: challenges/arcade/deadline-lab/writeup.md

Double Entry

IDEMPOTENCY

EVENT REPLAY

Replay delivered accounting events exactly once per event ID.

EVIDENCE AND CALCULATION

The 18 delivered rows represented 14 unique events. IDs 12, 4, 8 and 0 were duplicates and were ignored on second delivery. Unique charges totaled 11,559 cents; unique refunds totaled 9,134 cents.

```
opening + charges - refunds  
27,761 + 11,559 - 9,134 = 30,186
```

```
python challenges/arcade/double-entry/scripts/solve.py  
# unique_events=14  
# final_balance_cents=30186
```

VERIFIED RESPONSE

```
answer 30186 → {"ok":true,"flag":"flag{helpdesk_dc43606549c7d101}"}
```

Codex translated the stated event-identity rule into an idempotent replay and checked both subtotals. Playwright/CDP obtained the original export and verified the integer-cent balance. Artifact SHA-256: 7d6415b1...9e0b89c3.

Individual record: challenges/arcade/double-entry/writeup.md

Beacon Six

WAV

MORSE TIMING

Recover a six-digit identifier from an 8.56-second keyed-tone acquisition.

ANALYSIS

The mono 8 kHz PCM signal was not DTMF: standard dual-frequency energy tests failed. Energy runs instead revealed an 80 ms Morse unit. One active unit was a dot, three were a dash, and three silent units separated digits.

```
-....  -...-  -...-  -....  -...-  -....  
  6      0      9      6      3      7
```

```
python challenges/arcade/beacon-six/scripts/solve.py  
# groups=-....  -...-  -...-  -....  -...-  -....  
# answer=609637
```

VERIFIED RESPONSE

```
answer 609637 → {"ok":true,"flag":"flag{voicemail_d94ada40a244926c}"}
```

Codex supported waveform triage and the dependency-free timing decoder; the incorrect DTMF hypothesis was explicitly rejected. The WAV was downloaded byte-for-byte through the authorized browser session. SHA-256: 4dfdf92a...a73f2c9.

Individual record: challenges/arcade/beacon-six/writeup.md

Terminal Echo

CURSOR STATE

OVERWRITE

Render a cursor-control stream on an initially blank eight-column line.

EXECUTION

Only three operations existed: ESC[nG moved to one-based column n, backspace moved left without erasing, and printable characters overwrote the current cell. Every segment wrote a lowercase decoy, backspaced and overwrote it with uppercase.

```
ESC[1G a BS A → column 1 = A
ESC[2G m BS E → column 2 = E
...
final columns 1..8 = A E Y I O A K N

python challenges/arcade/terminal-echo/scripts/solve.py
# answer=AEYIOAKN
```

VERIFIED RESPONSE

```
answer AEYIOAKN → {"ok":true,"flag":"flag{printq_0201cfbd05f9c743}"}
```

Codex implemented the narrow terminal state machine with explicit failure for unsupported escapes; the final cell state was checked manually. Artifact SHA-256: 7ea7d3e3...d7a94a3d.

Individual record: challenges/arcade/terminal-echo/writeup.md

Custodian Records

GF(257)

LAGRANGE

Recover $f(0)$ from three points on a degree-at-most-two polynomial over the supplied prime field.

CALCULATION

The shares were $(1, 15)$, $(3, 198)$ and $(5, 16)$. Lagrange basis coefficients at zero were 34, 63 and 161. Their field contributions were 253, 138 and 6.

$$f(0) = (253 + 138 + 6) \bmod 257 = 140$$

```
python challenges/arcade/custodian-records/scripts/solve.py
# answer=140
```

VERIFIED RESPONSE

```
answer 140 → {"ok":true,"flag":"flag{vending_48e272382e88a354}"}
```

Codex implemented finite-field interpolation; every non-zero denominator and intermediate coefficient was printed for audit. Artifact SHA-256: cfa870fc...3610edb.

Individual record: challenges/arcade/custodian-records/writeup.md

Switchboard

XOR GRAPH

CONSTRAINT PROPAGATION

Recover binary states A through H from one anchor and same/different relationships.

REASONING

A relationship value 0 meant equality and 1 meant difference, equivalent to XOR. Starting with A=0, graph propagation produced:

```
A=0 B=1 C=1 D=1 E=0 F=0 G=1 H=1
answer=01110011
```

```
python challenges/arcade/switchboard/scripts/solve.py
```

The solver added both directions of every relationship and rejected any contradiction with an already assigned bit.

VERIFIED RESPONSE

```
answer 01110011 → {"ok":true,"flag":"flag{legacy_b81bde2dc1e65caf}"}
```

Codex built the propagation and consistency check; all seven equations were also checked manually. Artifact SHA-256: ac01fa68...c9e6bdfc.

Individual record: challenges/arcade/switchboard/writeup.md

Device Messages

SHARED PRIME

TEXTBOOK RSA

Decrypt the first device's raw RSA ciphertext after discovering shared key material.

CRYPTOGRAPHIC FAILURE

The two moduli shared prime 1051:

```
gcd(1123519, 1060459) = 1051
1123519 = 1051 × 1069
 $\phi(n_1)$  = 1,121,400
 $d = 65537^{-1} \bmod \phi(n_1) = 766,673$ 
 $m = 342096^{766673} \bmod 1123519 = 760422$ 

python challenges/arcade/device-messages/scripts/solve.py
```

The root cause is duplicated prime material from weak or repeated randomness. Because the task explicitly used textbook RSA without padding, modular exponentiation recovered the decimal plaintext directly.

VERIFIED RESPONSE

```
answer 760422 → {"ok":true,"flag":"flag{entropy_4f4121eb24ad9bce}"}
```

Codex proposed the batch-GCD check and implemented the transparent factor/totient/inverse steps; all values were locally recomputed. Artifact SHA-256: 1efd057c...5d153332.

Individual record: challenges/arcade/device-messages/writeup.md

Afterlife

CONWAY

THREE GENERATIONS

Advance a 6×6 Conway board three simultaneous generations with dead boundaries.

EXECUTION RECORD

A second grid buffered every generation so newly calculated cells could not affect neighbors in the same step. The third generation was:

```
011000
001000
101011
001111
110000
000000
```

```
python challenges/arcade/afterlife/scripts/solve.py
# answer=011000001000101011001111110000000000
```

VERIFIED RESPONSE

```
answer 011000001000101011001111110000000000 → {"ok":true,"flag":"flag{iris_63dcc5c2ad77ec1d}"}
```

Codex implemented explicit dead-boundary neighbor counting and printed all three intermediate boards. An independent calculation matched the final 36 bits. Artifact SHA-256: f4460afc...765be66.

Individual record: challenges/arcade/afterlife/writeup.md

Required four-point coverage matrix

ROOT CAUSE REPRODUCIBLE POC FIX / MITIGATION AI USAGE

Every individual `writeup.md` now ends with a standardized “Required submission summary” using the four exact headings requested by the platform. The matrix below is a concise index; commands, evidence and verifier output remain in the linked challenge records.

Challenge	Root cause / solving invariant	Reproducible PoC	Fix / mitigation	AI validation
Recon	Broken provenance and object binding	Replay custody; hash exact manifest	Immutable object/revision approvals	Visual, metadata and source checks
The Portal	Mutable reference reused after approval	Pause, mutate, resume, dispatch	Deliver approved immutable version/digest	403 control and delivered-version proof
The Insider	Subjects counted instead of people	Replay state; submit two realm tickets	Quorum on immutable person identity	Signed payload and employee-number checks
The Evidence	Timing channel plus residual profiles	Decode two lanes; CRC, XOR, XZ	Signed collectors, secure purge, monitoring	Calibration/ECC/hash/verifier controls
The Cipher	AES-GCM nonce reuse	Recover H; forge release tuple	Atomic nonce state and key rotation	Both source tags and live validator
The Tunnel	Cursor fields not integrity-bound together	Rebind generation in fresh TLS session	Opaque authenticated cursor/handle	Offline trace regeneration and EXPORT proof
The Implant	Policy flags audited but not enforced	Map unique event chain to context 35	Pre-load/connect enforcement; secret hygiene	CRC, pointers, GCM and endpoint checks
Nightjar’s Nest	Queued raw pointer without strong reference	Trigger UAF; derive arbitrary read/write	Stable references and checked handles	Fresh ASLR/state and board acceptance
Epilogue	Unauthorized module DNS-transfer chain	Run exact SQL on training and held-out sets	Trust, deployment and egress enforcement	Confusion matrix and independent evaluator
Route Zero	Directed shortest-path invariant	Dijkstra returns cost 49	Validate direction and nonnegative weights	All paths plus official verifier
Pixel Ledger	Two-axis run constraints	Enumerate 2^{16} grids; assert unique	Validate clues and uniqueness	Manual clues plus official verifier
Deadline Lab	Weighted completion with precedence	Enumerate all 720 schedules	Validate DAG, objective and tie-break	Complete feasible search
Double Entry	Duplicate event delivery	Replay once per ID; balance 30186	Atomic idempotency and unique ID	Counts, subtotals and verifier
Beacon Six	Morse timing, not assumed DTMF	Decode 80 ms tone units	Format/timing/noise validation	Rejected DTMF hypothesis; verifier

Challenge	Root cause / solving invariant	Reproducible PoC	Fix / mitigation	AI validation
Terminal Echo	Cursor overwrite semantics	Emulate documented controls	Bounded explicit terminal state machine	Manual cell trace and verifier
Custodian Records	Interpolation in GF(257)	Lagrange at x=0 returns 140	Reviewed field math; authenticate shares	Intermediate bases and verifier
Switchboard	Bidirectional XOR relationships	Propagate anchor through graph	Reject inconsistent/unanchored components	All equations and verifier
Device Messages	RSA moduli share prime 1051	GCD, factor, derive d, decrypt	Healthy RNG, batch GCD, OAEP	All crypto values and verifier
Afterlife	Simultaneous-generation invariant	Double-buffer three Life steps	Explicit boundaries and generation tests	Intermediate boards and verifier

Audit result. Nineteen write-ups checked; all nineteen include the exact headings *Root cause*, *Reproducible PoC*, *Fix / mitigation*, and *AI usage*. Standalone puzzles explicitly state when no service vulnerability exists and provide correctness safeguards rather than inventing remediation.

Outstanding questions and defensive follow-up

Challenge status. There are no unresolved challenges in the event scope represented by this repository. All nine chain/branch/finale stages and all ten Independent Arcade puzzles have acceptance proof: 19/19 solved. Accordingly, there is no unverified flag candidate presented as an open lead.

The following leads remain useful for defensive validation and hardening, but are not required to establish the submitted solves:

Area	Open lead	Recommended next check
Portal	Other delivery states may reopen mutable references.	Add mutation tests at queued, waiting, resumed and partial-cursor states.
Insider	Other realms may create additional aliases for one employee.	Deduplicate on immutable person ID across every identity provider.
Evidence / Cipher	Timing channels could use non-binary phase or rank alphabets.	Profile inter-arrival distributions and rank permutations across all sensors.
Tunnel	Other opaque-looking tokens may duplicate authority fields.	Fuzz recombination of reference, generation, purpose and session fields.
Implant	Twenty-nine destination-only violations remain suspicious.	Correlate them with independent module trust and deployment history.
Nest	Other queue cancellation/completion paths may retain stale pointers.	Instrument strong-reference transitions and allocator lifetime tests.
Epilogue	Production data may contain clock gaps or ambiguous overlapping leases.	Fail closed or surface an explicit "unattributable" class instead of guessing.

FINAL ASSESSMENT

The operation’s repeated design failure was broken binding: approval separated from bytes, account subject separated from human identity, cursor reference separated from generation, authorization flags separated from enforcement, and queued work separated from object lifetime. The successful investigation repeatedly restored those bindings through exact state reconstruction and validation.

Callsign: **rupesh**
Investigator submission

Date: **27 September 2026**
Final transmission / v1.0

Evidence, analysis and script map

The repository is the reproducibility package. The following index identifies the primary entry points; challenge write-ups contain the full artifact hashes and intermediate values.

Challenge	Primary retained evidence	Key reproducibility material
Recon	assets/mail/: EML, WAV, badge, three facility photographs, custody JSONL	writeup.md contains replay and manifest scripts; original PNG hashes retained
The Portal	assets/portal-support/, portal-client.js	analysis/portal-job-11-delivery.txt, acquisition receipt and handover JSON
The Insider	workflow.db, directory.json, deployments.json	Deterministic journal replay and browser-console reproduction in writeup.md
The Evidence	bundle.zip, workstation image, PCAP, historian DB, calibration and ELF files	Ten focused scripts under scripts/; recovered lanes, DB and document under analysis/
The Cipher	Two acquisition ZIPs, receiver PCAP, registrations, sensors and custody notes	PCAP extractor, transport/journal recovery, GCM probes and final forge under scripts/
The Tunnel	Stripped client, shared library, ABI and two laboratory captures	scripts/solve_tunnel.py with offline verification and fresh TLS solver
The Implant	ELF core, stripped validator, acquisition notes and Tunnel handover	Core mapper, event parser, context solver and submit helpers under scripts/
Nightjar's Nest	Stripped handler	probe_nest.py, solve_nest.py, and controlled flag submission helper
Epilogue	epilogue-training.json	epilogue_analyze.py, exact detector SQL, local SQLite and evaluator result JSON
Independent Arcade	Ten exact JSON/WAV puzzle artifacts under challenges/arcade/*/assets/	One write-up and one deterministic solver per puzzle; aggregate verifier at challenges/arcade/verify_arcade_answers.js

Selected integrity anchors

Artifact	SHA-256
Portal restricted delivery	3640a5148aa1a60b8293aac052f0c4fd93ebcd8b0be3f1ed2b174c1a8a59c640
Evidence bundle ZIP	84bc1b7c9a5da42d58b2479c2933f7843585ef97cbad6312f95a4ee7e3b2a968
Recovered Evidence document	4484648495b0dab0b48e533c638cdc5224f5d86d7bedbfbf8fe4cf88283681de
Cipher acquisition ZIP	32a85ead76f37959fad2fcdfe907426969895a7032f8bc7c71a1c327ef64193e
Tunnel maintenance client	a380bed19cb86766d19ecbd35ba24fef85ba060ae6084b85f9f53bd84dbdfcc4
Nightjar handler	c670dc665e81e251965af8623083cc89fa812cc0c8083fd85506a777793d2030
Epilogue training acquisition	6b8ea80305398c6eff619920b3104818b4f86c64b5ff66bcc9920b5cab5e8712

Limitations and interpretation

- Session-specific addresses, tokens, field credentials and randomized state are evidence examples, not reusable constants.
- Flags are included because this is the final competition dossier; they do not replace the execution and proof record.
- Browser snapshots and local profile state are supporting acquisition traces. The authoritative proof remains retained response bytes, hashes, scripts and service acceptance.
- Detailed remediation, background explanations and complete code blocks remain in each stage write-up to keep this consolidated PDF navigable.

End of dossier. Evidence preserved. Operation reconstructed. Verification trail retained in the accompanying repository.